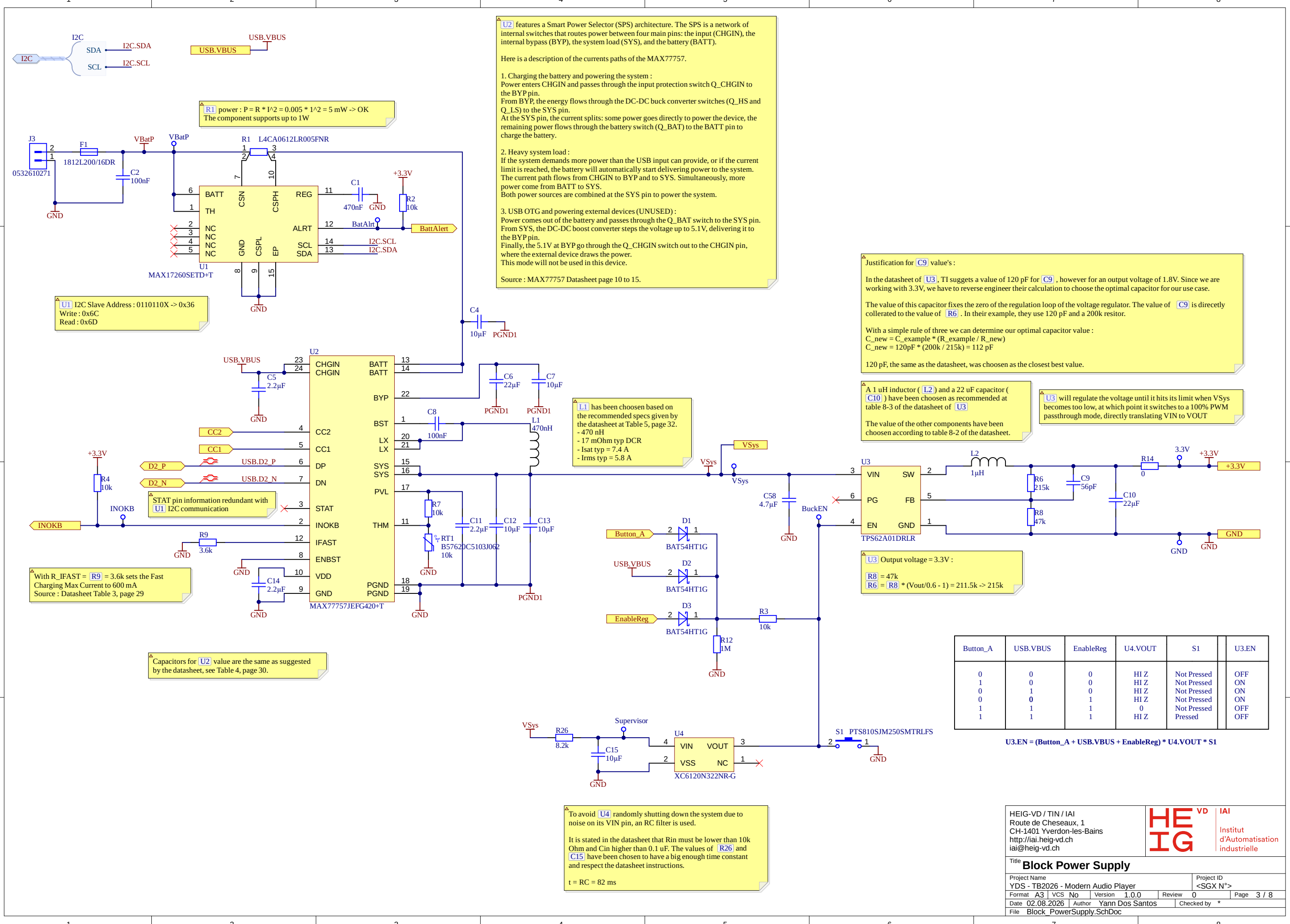
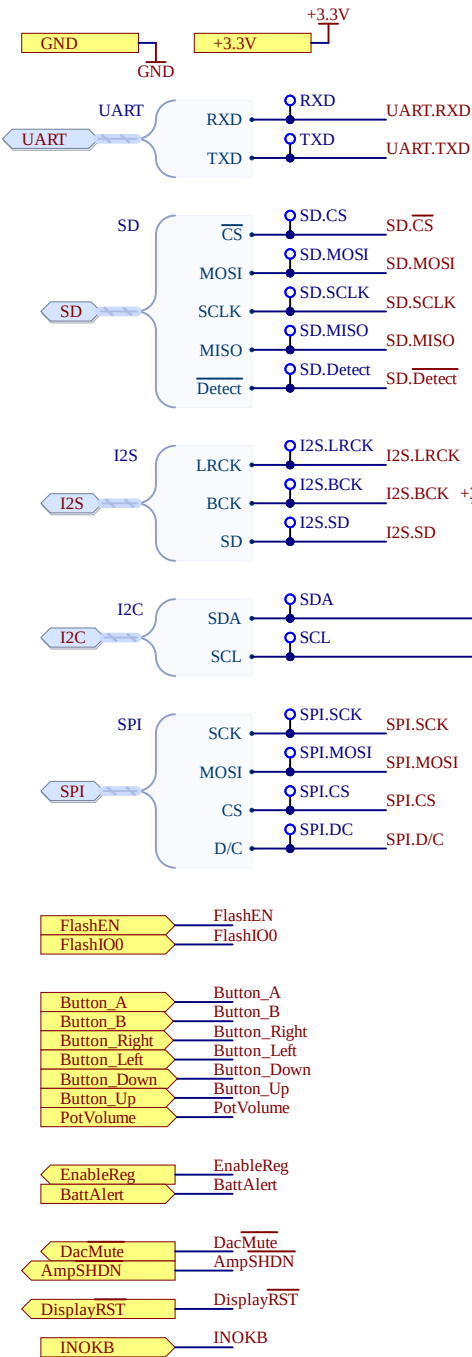






I2C 400 kHz, pull-up resistor calculation.
Source : <https://www.ti.com/lit/an/slva689/slva689.pdf?ts=1778213985956>

$R_{p(min)} = (V_{cc} - 0.4) / 0.003 = (3.3 - 0.4) / 0.003 = 967 \text{ Ohm}$
Rp(min) is fixed no matter other parameters (since we only use +3.3V)

$R_{p(max)} = t_r / (0.8473 * C_b)$
tr is the maximum rising time depending of the frequency of the communication
Cb is all the capacitance of the bus combined. To determine it, we add all the pin input capacitance of each device connected to the I2C bus, and then take a safety margin.

$t_r = 300 \text{ ns}$ (see source Table 1)

To calculate Cb, we must add the capacitance of U10, U11, U8 and U1
Cb = 28 pF

Now we take a safety margin and calculate R (we will use C = 50 pF and C = 100 pF) :
 $R_{p(max)1} = 300\text{ns} / (0.8473 * 50\text{pF}) = 7.08\text{k Ohm}$
 $R_{p(max)2} = 300\text{ns} / (0.8473 * 100\text{pF}) = 3.54\text{k Ohm}$

We these calculation, 2k2 is a good choice for the pull-ups resistors

Pin 4 to pin 7 are input only. (SENSOR_VP, SENSOR_VN, GPIO34, GPIO35)
See ESP32-WROVER-E datasheet Table 3, page 11

For the potentiometer controlling the volume, it must be connected to ADC1 (GPIO32 to GPIO39). ADC2 is blocked when the Wi-Fi is used. (Source : ESP32 Technical Reference Manual, page 741)

In the code, an attenuation must also be configured (Atten=3) to use a bigger linear range from 150mV to 2450mV. Resistor and R67 iare used to cap the max voltage to ~2518 mV and min voltage to ~147 mV
(Source : ESP32 Datasheet, Table 4-4, page 45)

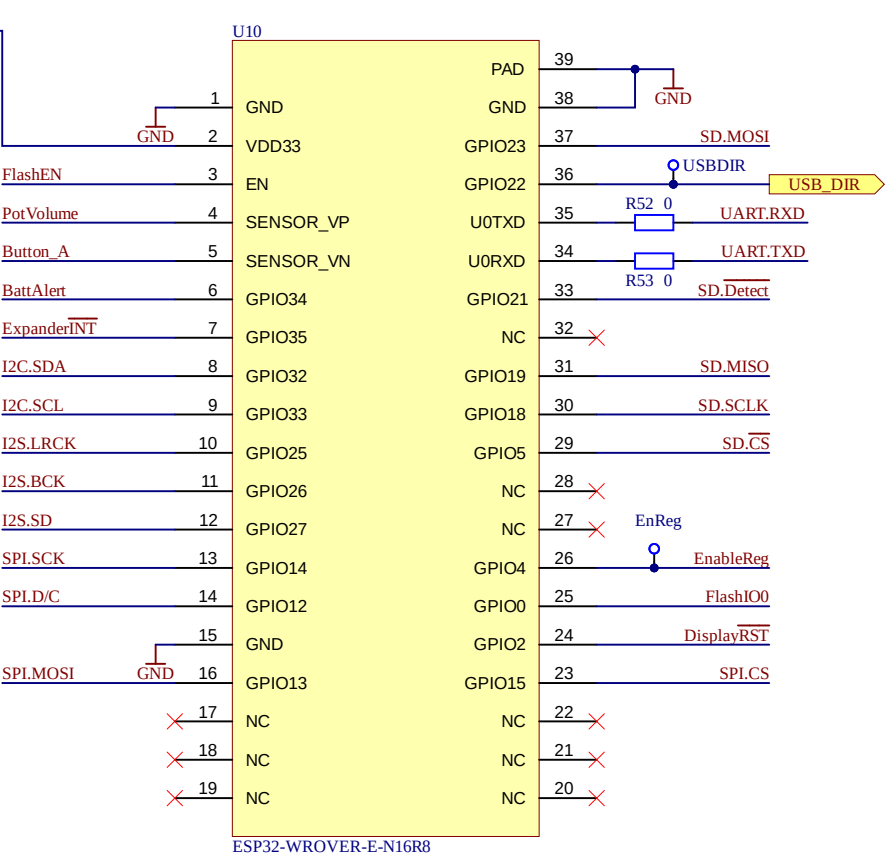
Signal -> Physical Pin (Internal connection)

PotVolume -> SENSOR_VP (ADC1_CH0)

Button_A -> SENSOR_VN (RTC_GPIO3)

BattAlert -> GPIO34 (RTC_GPIO4)

ExpanderINT -> GPIO45 (RTC_GPIO5)



Thes ESP32 features a GPIO Matrix allowing each pin to be connected to any internal peripheral signal (Technical Reference Manual page 115).

However, for better high speed communication performance, direct connections for peripherals will be used unless impossible.
See Table 6.10-1, page 133, Technical Reference Manual

GPIO0, GPIO2, GPIO5, GPIO12 and GPIO15 are strapping pins used during the boot sequence of the ESP32, modifying their default values can make the chip freeze or crash. (ESP32 Datasheet, Table 3-1, page 22)

It is best to leave these pins unconnected or use them as output, since these pins only require a specific value during the boot sequence.

DISPLAY SPI :
Since the display uses half-duplex SPI mode (no MISO) so no inputs towards the ESO32, it will be connected using the HSPI peripheral as it has the most strapping pins

SPI.SCK -> GPIO14 (HSPICLK)
SPI.MOSI -> GPIO13 (HSPID)
SPI.CS -> GPIO15 (HSPICS0)
SPI.D/C -> GPIO12
DisplayRST -> GPIO2

SD CARD SPI :
HSPI being used by the display, the other free SPI peripheral is VSPI

SD.CS -> GPIO5 (VSPICS0)
SD.MOSI -> GPIO23 (VSPID)
SD.SCLK -> GPIO18 (VSPICLK)
SD.MISO -> GPIO19 (VSPIQ)
SD.Detect -> GPIO21

SD.CS is connected to the GPIO5 strapping pin with a pull-up resistor but it is not an issue since the default value of this pin is 1, it will not cause any problem for the boot sequence. (ESP32 Datasheet, Table 3-1, page 22)

USB-UART Bridge :
UART.RXD -> U0TXD
UART.TXD -> U0RXD

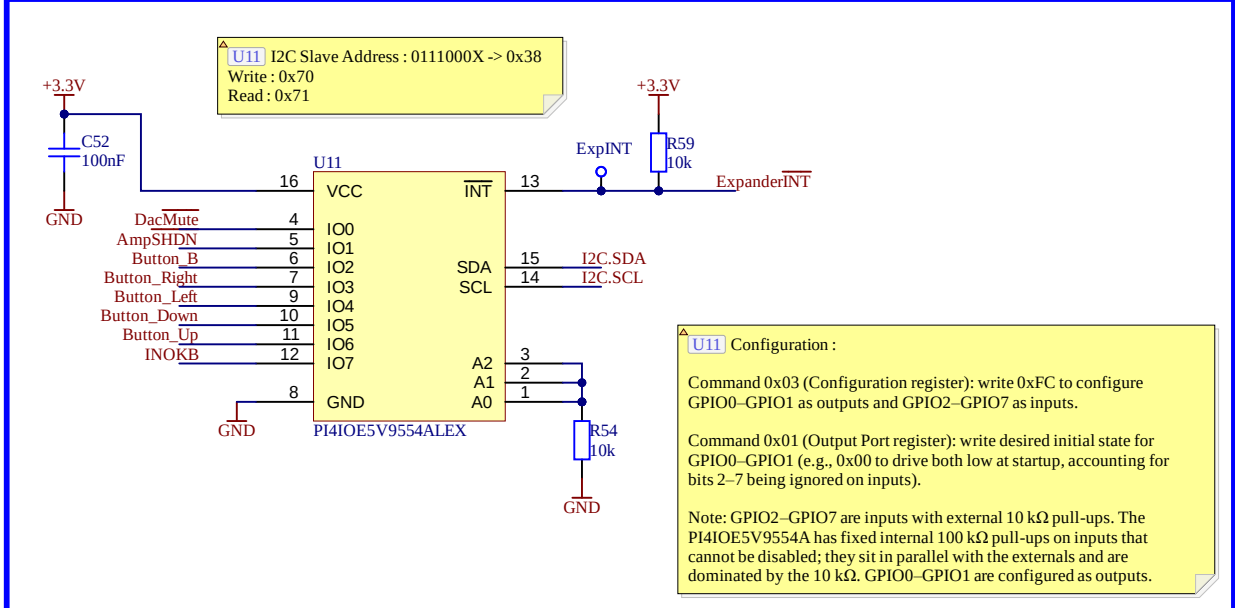
I2C Interface :
The I2C peripheral inside the ESP32 has no direct connection pin, so it can be connected to any GPIO using the GPIO Matrix. See ESP32 Datasheet section 4.8.4

I2C.SDA -> GPIO32
I2C.SCL -> GPIO33

DAC I2S :
The I2S peripheral inside the ESP32 has no direct connection pin, so it can be connected to any GPIO using the GPIO Matrix. See ESP32 Datasheet section 4.8.5

I2S.LRCK -> GPIO25
I2S.BCK -> GPIO26
I2S.SD -> GPIO27

I/O Expander

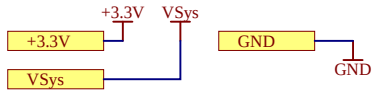


U11 Configuration :

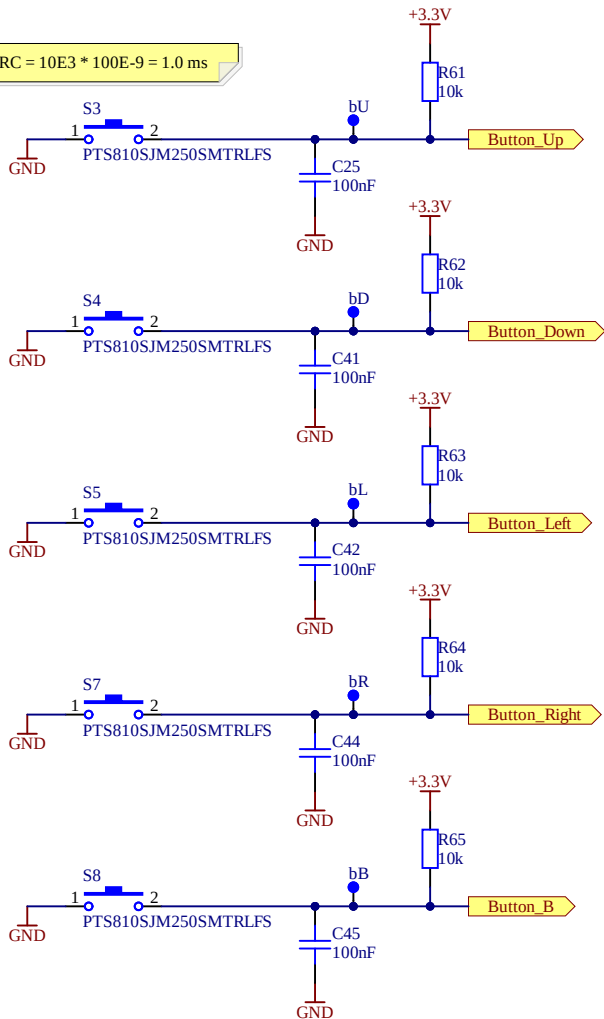
Command 0x03 (Configuration register): write 0xFC to configure GPIO0–GPIO1 as outputs and GPIO2–GPIO7 as inputs.

Command 0x01 (Output Port register): write desired initial state for GPIO0–GPIO1 (e.g., 0x00 to drive both low at startup, accounting for bits 2–7 being ignored on inputs).

Note: GPIO2–GPIO7 are inputs with external 10 kΩ pull-ups. The PI4IOE5V9554A has fixed internal 100 kΩ pull-ups on inputs that cannot be disabled; they sit in parallel with the externals and are dominated by the 10 kΩ. GPIO0–GPIO1 are configured as outputs.

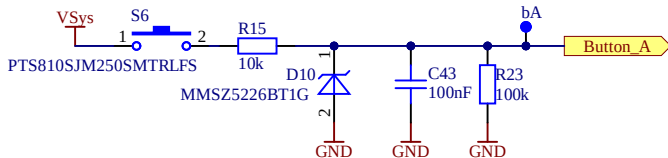


$\tau = RC = 10E3 \times 100E-9 = 1.0 \text{ ms}$



Since **S6** (Button A) is also used as the Power ON button, it is connected to VSys instead of +3.3V, as the +3.3V is only active once the system has been powered on.

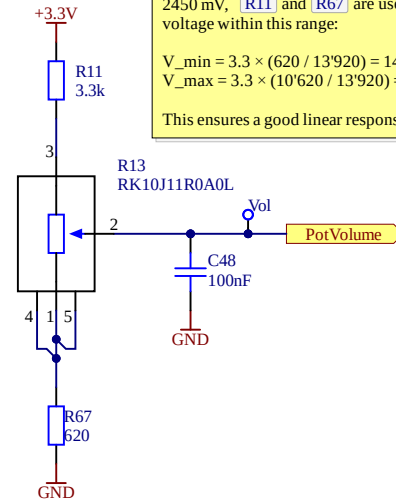
To protect the ESP32 GPIO input, a 3.3V Zener diode is used.



Since the ESP32 linear ADC range is from 150 mV to 2450 mV, **R11** and **R67** are used to keep the wiper voltage within this range:

$V_{\min} = 3.3 \times (620 / 13'920) = 147 \text{ mV}$
 $V_{\max} = 3.3 \times (10'620 / 13'920) = 2.518 \text{ V}$

This ensures a good linear response of the volume knob



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Title

Project Name
YDS - TB2026 - Modern Audio Player

Project ID
<SGX N°>

Format A4 | VCS No | Version 1.0.0 | Review 0 | Page 6 / 8

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